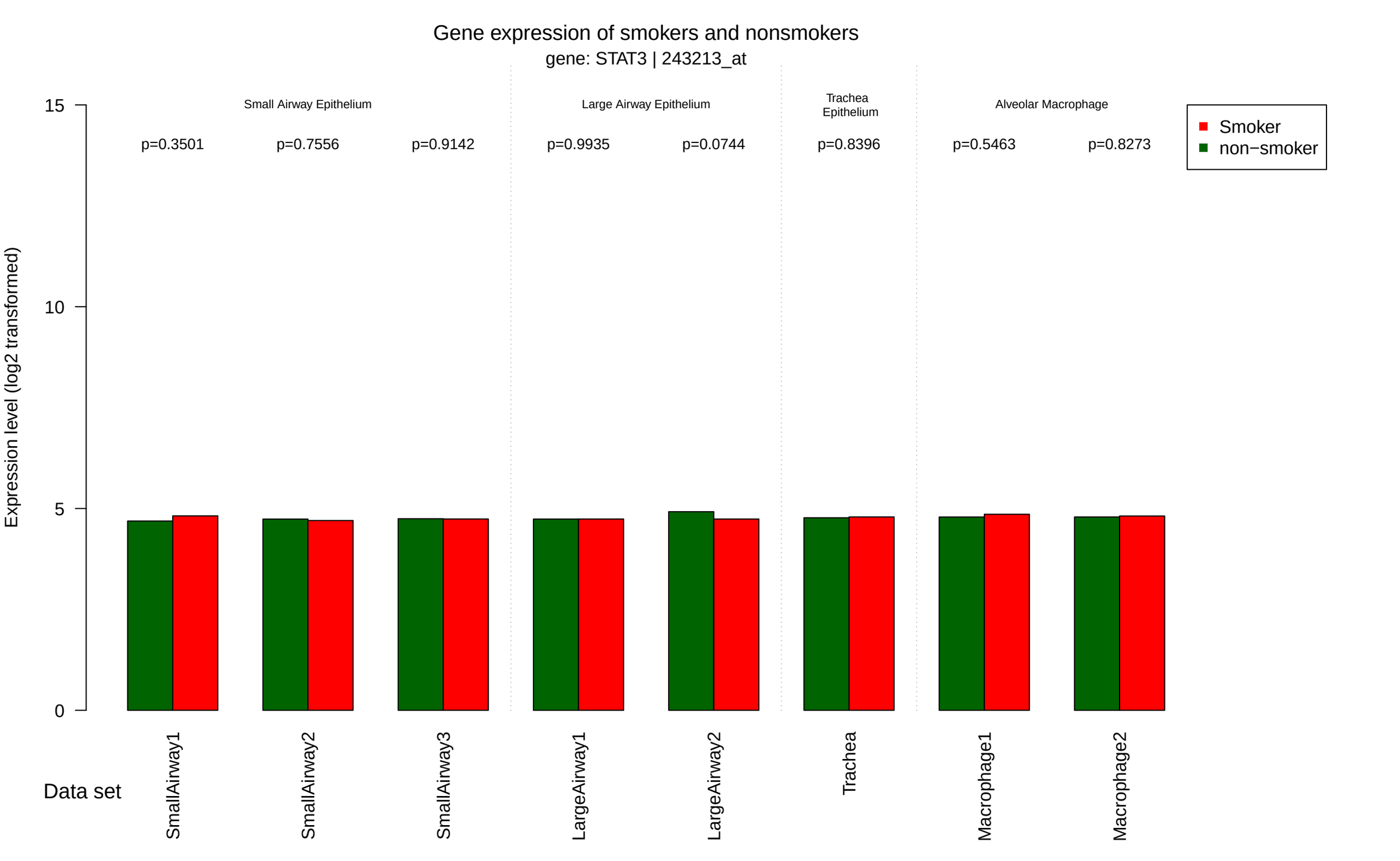
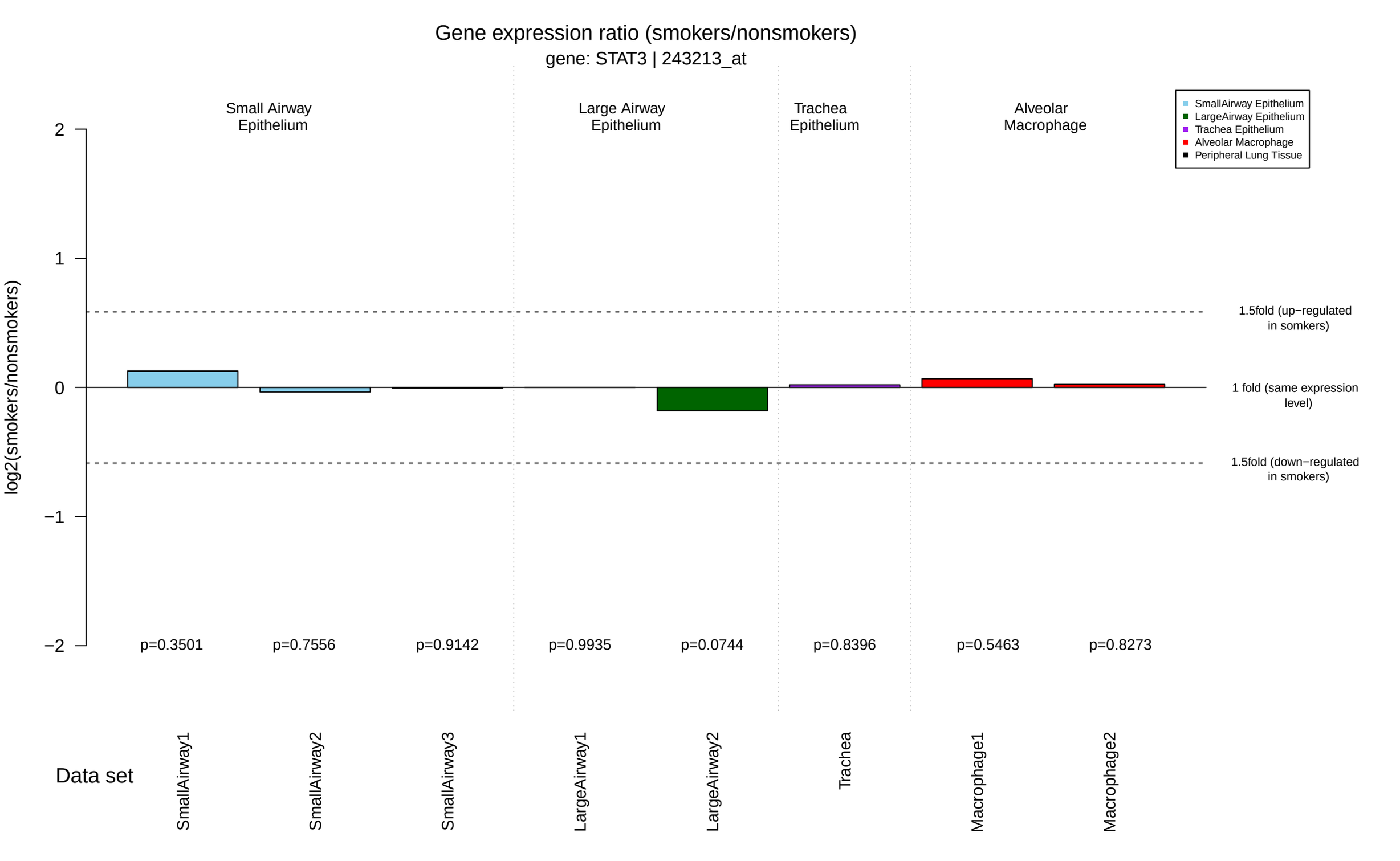
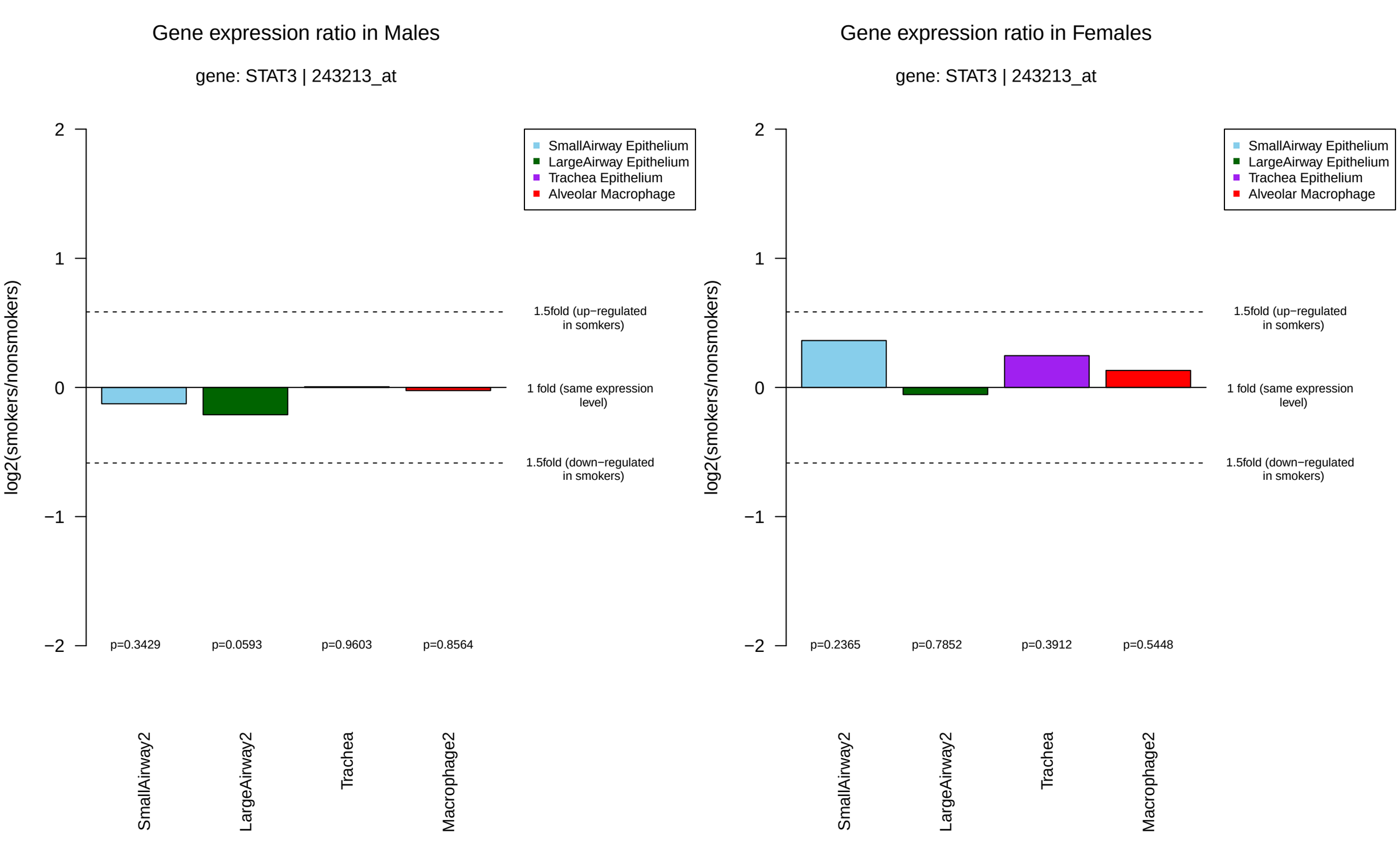
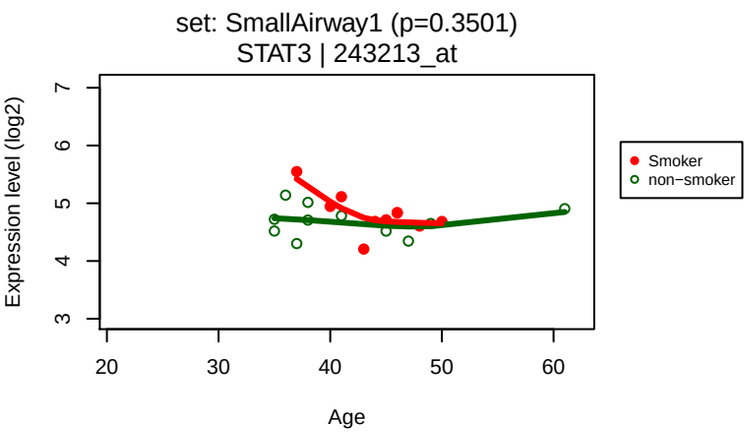
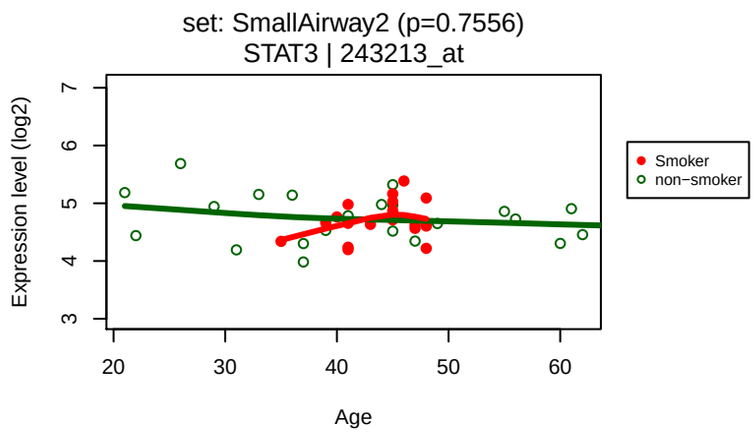
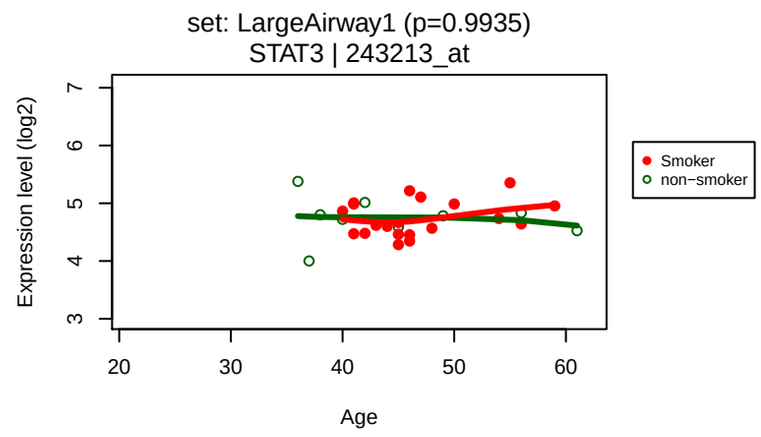

Gene expression of smokers and nonsmokers
gene: STAT3 | 243213_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3501
p=0.7556
p=0.9142
p=0.9935
p=0.0744
p=0.8396
p=0.5463
p=0.8273
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

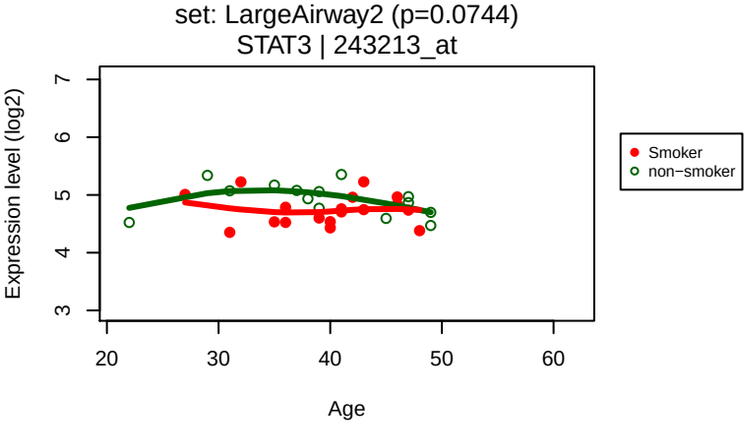

Gene expression ratio (smokers/nonsmokers)
gene: STAT3 | 243213_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3501
p=0.7556
p=0.9142
p=0.9935
p=0.0744
p=0.8396
p=0.5463
p=0.8273
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

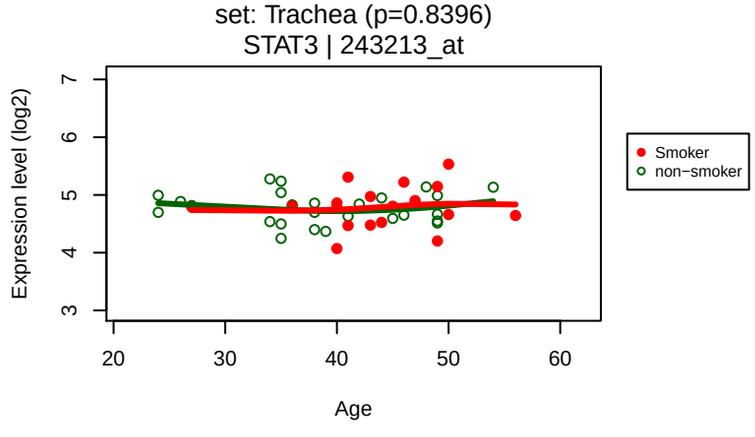

Gene expression ratio in Males
gene: STAT3 | 243213_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3429
p=0.0593
p=0.9603
p=0.8564
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAT3 | 243213_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2365
p=0.7852
p=0.3912
p=0.5448
SmallAirway2
LargeAirway2
Trachea
Macrophage2

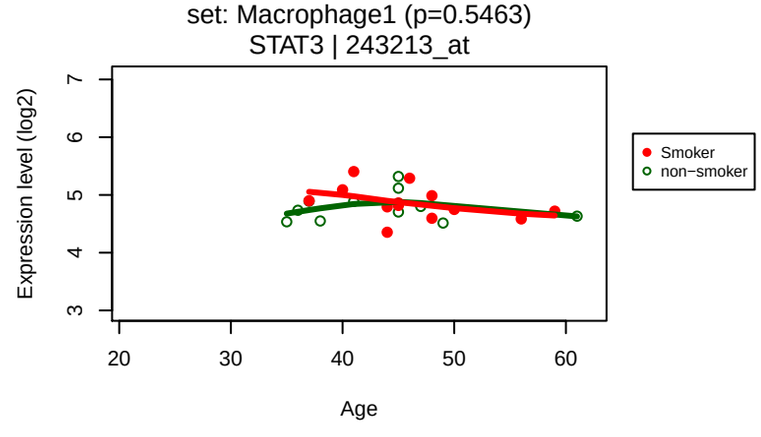

set: SmallAirway1 (p=0.3501)
STAT3 | 243213_at
Expression level (log2)
Age
Smoker
non−smoker

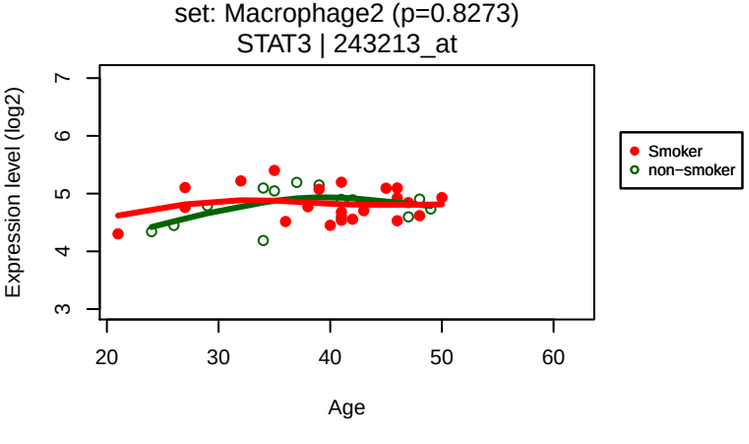

set: SmallAirway2 (p=0.7556)
STAT3 | 243213_at
Expression level (log2)
Age
Smoker
non−smoker

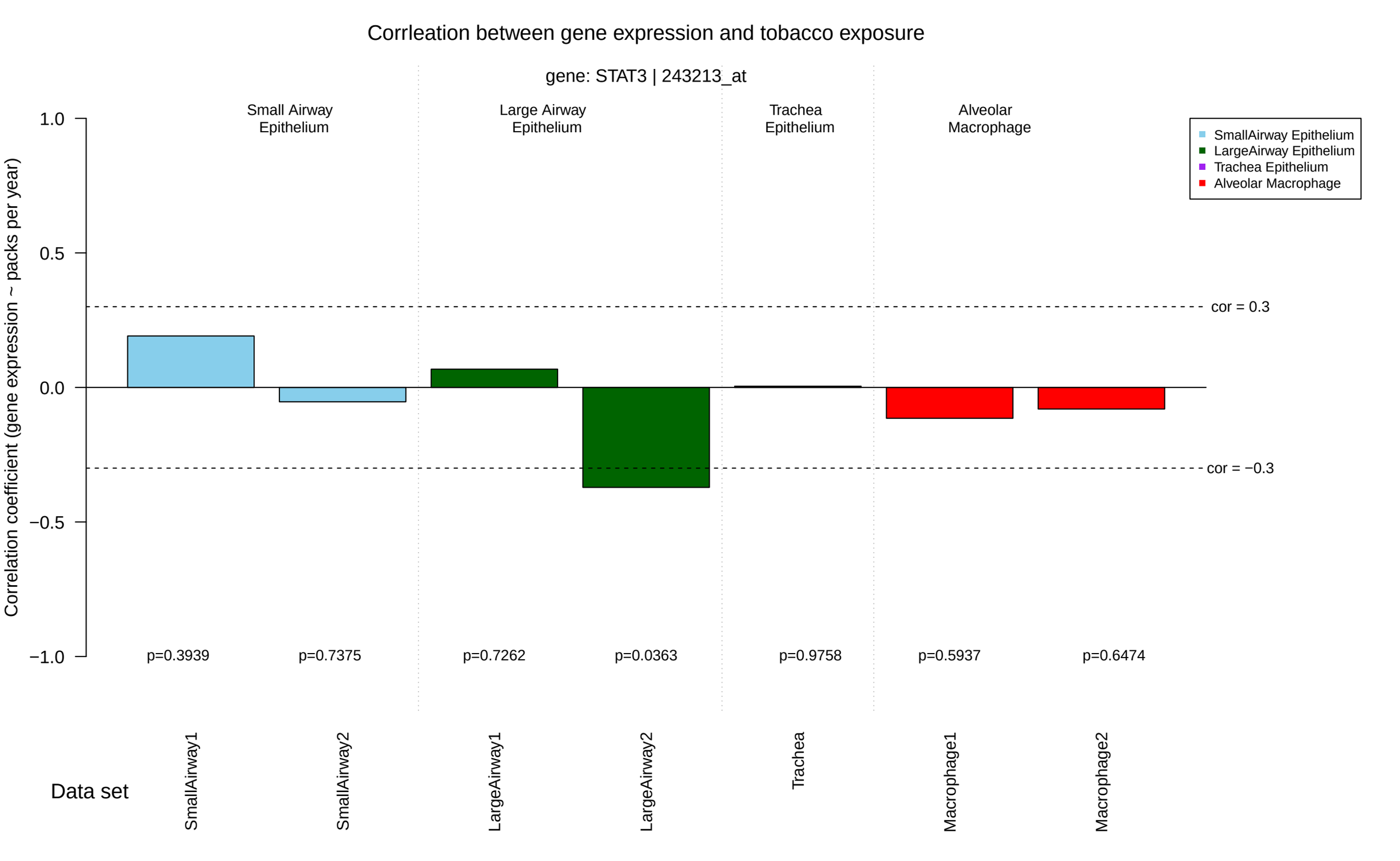

set: LargeAirway1 (p=0.9935)
STAT3 | 243213_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0744)
STAT3 | 243213_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8396)
STAT3 | 243213_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5463)
STAT3 | 243213_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8273)
STAT3 | 243213_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAT3 | 243213_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3939
p=0.7375
p=0.7262
p=0.0363
p=0.9758
p=0.5937
p=0.6474
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2